The Conceptual on the Impact of Digital Competence on

Non-academic Staff's Job Performance in the Vietnam's Higher Educational Context

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ABSTRACT

Amid a digitally growing world, digital competence has emerged as a critical concept in discussing what kinds of proficiency individuals should have. It is believed that, in addition to professional competence, digital competence is required for the workforce to improve their work results when much of their daily work uses technologies. Nonetheless, a high rate of the Vietnamese force faces a low level of digital competence, which aligns with the global trend. Meanwhile, previous studies have mainly concentrated on developing digital competence frameworks and proposing roadmaps to apply them; this current study may help seek answers to the questions of how the digital competence of the non-academic staff impacts their job performance. This proposed study may use survey quantitative research design to serve descriptive and evaluative research purposes based on the snowball sampling techniques for data collection. This study could be a fitting response to explain to what extent digital competence impact job performance among the non-academic staff in Vietnam universities by surveying full-time members from a private university. In addition, the findings from this study may assist educational organizations in selecting strategies for improving job performance based on a better understanding of the impact of digital competence on it.

CCS CONCEPTS

• **Social and professional topics Professional topics Computing education Computing literacy**;

KEYWORDS

digital competence, non-academic staff, the UTAUT model, job performance

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1 INTRODUCTION

Accelerated technological developments have revolutionized society and continue impacting and changing how organizations work [1]. In line with this development, digital transformation has become a trend and should be viewed as a lifeblood for the success of nations and organizations. In the education industry, digital transformation blew a new mind into educational organizations, especially in higher educational institutions. Nonetheless, this situation also forces educational organizations to simultaneously apply different solutions for strengthening their inner strength to enable them to gain a head start by taking a shortcut in the digital transformation race. In the context of the growing implementation of digital transformation globally, adopting the appropriate digital technologies and nurturing the workforce competencies to apply technologies properly become a wise strategy. Developing an elite workforce, thus, with adequate digital competence, is deemed the key to the success of digital transformation for organizations [2], and digital competence has become one of the core competencies that organizations need to equip their workforce to create a sustainable competitive advantage. Because it is argued that it is not cutting-edge technology that creates a competitive advantage for organizations; instead, people with the necessary capabilities will make the difference [3].

Notwithstanding the vital role of digital competence mentioned above, the global workforce faces an outstanding problem: the low level of digital competence [4]. Even the EU, considered the region with the fastest digital conversion speed [5], faces this problem. More than 75 million persons in the EU who are employed or searching for work, account for 35 % of the EU labor force, lack even the most fundamental digital skills [6]. A similar trend was observed in Vietnam, a developing country in Southeast Asia. The lack of digital competence among the Vietnam labor force is even more evident. According to Trinh (2022), most Vietnamese workers lack core work skills, especially digital ones. In other words, there is a massive gap between employee digital competence and employer expectancy [7]. In education, the research conducted by British Council Viet Nam (2022 denoted that an overall positive picture emerges from the survey results regarding the readiness of Vietnam's HEIs in digital transformation, notable achievements in educational management, and innovation in teaching and learning. Nevertheless, the low level of digital competence is deemed one of the most significant challenges for HEIs. It is argued that a low

level of digital competence workforce severely hinders Vietnam in the digital transformation process both at the competitive capability of national and organizational levels and well-being [8], job satisfaction [9], and work engagement [10] at the individual level.

In addition to the problem of the workforce's digital competence, Vietnamese workers face difficulties with job performance. Although work productivity is a narrower concept than work performance [11] and not the same as job performance [12], the statistical indicators related to productivity may partly reflect the level of job performance among the Vietnamese labor force. Despite the reasonably high growth attained in the recent decade, Vietnam's labor productivity remained in the group of least productive Asian nations in 2021 (Asian Productivity Organization, 2021). It has been argued that employee job performance directly impacts an organization's success [13] and growth [14]. Job performance also benefits employees when it is an assessment scale for their recognition, promotion, career opportunities, and skill mastery [15]. Hence, poor job performance adversely affects both organizational and personal development when it can damage ripple effect across the entire business, resulting in a widespread loss of motivation and productivity and a decrease in customer satisfaction. Concerning education, teachers' performance has been demonstrated to affect students' achievements and learning [16].

Based on the above discussion, the proposed study aims to understand better the impact of the digital competence of non-academic staff in Vietnam universities on their job performance. Additionally, an empirical investigation will shed light on the mediation role of the determinants of the acceptance and usage of digital technology and self-efficacy in the relationship between digital competence and job performance.

The future study is framed to answer the following research questions in the Vietnamese higher education context:

- What is the digital competence of non-academic staff in Vietnam universities in relation to their current job performance levels?
- How do performance expectance, effort expectance, and social influence mediate the relationship between digital competence and job performance?
- How does self-efficacy mediate the relationship between digital competence and job performance?
- To what extent does the digital competence of non-academic staff in Vietnam universities directly impact their job performance?

2 LITERATURE REVIEW

2.1 Digital competence

As one of eight critical competencies for "Lifelong Learning and a "transversal key competence" which enables people to acquire other competencies [17], digital competence is conceptualized by many scholars to clarify the dimensions of this term. One of the widely accepted definitions is those of Ferrari (2012), who defined digital competence as a "set of knowledge, skills, attitudes (thus including abilities, strategies, values, and awareness) that are required when using ICT and digital media to perform tasks; solve problems; communicate; manage information; collaborate; create and share content; and build knowledge effectively, efficiently, appropriately, critically, creatively, autonomously, flexibly, ethically, reflectively for work, leisure, participation, learning, socializing, consuming, and empowerment" [18]. Although several terms are used interchangeably with digital competence, such as digital literacy and media literacy, digital competence is the most popular term used in the research. The researcher prefers to use digital competence in the current study for two reasons. First, digital competence is often employed as a broader concept [17], which suits the workplace context of the present study. Second, this term was conceptualized as a series of intertwined knowledge, skills, and attitudes, the three learning domains envisaged by [19], which can be measured.

Concerning digital competence frameworks, it must be mentioned about the chain of projects of researchers from the European Commission's Joint Research Centre because their definitions and frameworks were mainly used among scholars, especially in the education section [20]. In this study, the researcher will adopt the digital competence framework DigComp 2.1, developed by [21]. This framework divides digital competence into five competence areas: (i) Information and data literacy; (ii) Communication and Collaboration; (iii) Digital content creation; (iv) Safety; (v) Problem-solving.

2.2 Job performance

Job performance (interchangeably with work performance, employee performance) is a central construct in organizational psychology [22] and human resources management. It has become the primary concern of companies worldwide, especially in digital transformation, as technological innovations pose unprecedented competitive threats to organizations [23]. The concept and definition of job performance have received considerable scholarly attention over the past 30 years. There are two streams of the meaning of job performance; one focuses on the behavior aspect, and the other focuses on both behavior and outcomes. One of the widely accepted definitions of the behavioral aspect of job performance is that of Campbell (1990): "behaviors or actions that are relevant to the goals of the organization" [22]. This concise and succinct definition implies that work performance, a multidimensional construct, focuses on behavior rather than results with behavior related to the organization's goals [11]. Taking a different point of view, Roe (1999) denoted that behavior and outcome are two aspects of job performance; while the behavior aspect focuses on "*what*" employees do at work, the outcome aspect highlights the "*results*" of behavior [24]. Although some scholars presume job performance is a behavior rather than a result, these two aspects of job performance (behavior and outcome) are not mutually exclusive [25] and distinguishing between these two aspects can be challenging. Moreover, Sonnentag & Frese (2005), the two eminent scholars in human resource management, supported the perspective that performance comprises both a behavioral and an outcome aspect [26]. From the above justification, the researcher chooses the latter point of view of job performance in this proposed study.

Various research has been conducted on factors affecting job performance to explore its determinants. Al-Ahmadi, (2009) investigated whether and to what extent differences in employee demographics, job satisfaction, and organizational commitment influence performance. The findings revealed that organizational commitment, job satisfaction, and personal and professional variables are associated with job performance. Job satisfaction and

organizational commitment strongly predict nurses' performance [27]. Pandey (2018) identified and classified factors affecting job performance by conducting an integrative literature review. Based on the job demands-resources model (JD-R model) developed by Demerouti et al. (2001), Pandey divided these variables into two distinct categories, job demands and job resources, while the remaining variables, the researcher classified them into stressors. These three categories were generated into the JD-R-S framework, in which: (i) Job demands consist of physical, cognitive, and affective demands; (ii) Resources include personal resources, job resources, organizational resources, and social resources; (iii) Stressors encompass individual stressors, job stressors, and family stressors [25].

In terms of job performance assessment, it benefits both organizations and individuals. Regarding the organizational level, performance assessment is fundamental for personnel decisions or high-stakes appraisals and serves the development purpose. It is clear that the performance results according to feedback help to enhance and improve employee behaviors and outcomes. Regarding the individual level, performance assessment is a measure of self-appraisal for self-managed performance improvement [12]. There are three performance assessment methods: (a) performance ratings, which may perform by supervisors, peers, subordinates, or by workers themselves; (b) simulations which means individuals perform tasks in fabricated situations or with facsimiles of task materials; (c) technology-enhanced assessment, which based on technologies to enhance or replace other forms of performance assessment. It is noted that there is no best way to assess job performance; hence, the applicability of each method depends on the assessment purposes [12].

2.3 Theoretical framework, hypotheses development and conceptual framework

2.3.1 Theoretical framework. The researcher adopts the Unified Theory of Acceptance and Use of Technology (UTAUT) for this study (Figure 1). The UTAUT model was developed by Venkatesh et al. (2003) based on the integration and capture of the essential elements of eight previously established models, namely the Theory of Reasoned Action (TRA), the Technology Acceptance Model (TAM), the Motivational Model(TMM), the Theory of Planned Behavior (TPB), a combined TBP/TAM, the Model of PC Utilization, Innovation Diffusion Theory (IDT), and Social Cognitive Theory (SCT)). According to Venkatesh et al. (2003), performance expectancy, effort expectancy, social influence, and facilitating conditions are the four direct determinants of user acceptance and usage behavior. Performance expectance is defined as "the degree to which an individual believes that using the system will help him or her to attain gains in job performance". Effort expectancy is "the degree of ease associated with the use of the system". Social influence is the third determining factor with meaning is "the degree to which an individual perceives that important others believe he/she should use the new system". The last causal factor, facilitating conditions, is "the degree to which an individual believes that an organizational and technical infrastructure exists to support the use of the system" [28].

Along with the UTAUT model, this proposed study adopts the self-efficacy theory, indicating that self-efficacy impacts people's behavior [29]. In detail, people with a high level of self-efficacy view complex tasks as challenges to be mastered rather than threats to be avoided; conversely, people who doubt their capabilities shy away from complex tasks, which they view as personal threats. Self-efficacy also impacts task choice, effort, persistence, and achievement [30].

Moreover, the relationship between the two concepts, competence and performance, has drawn the great attention of scholars. According to the point of view of Campbell & Wiernik (2015), the association of competence and performance can be defined based on three choices: (i) a competency could refer to performance itself; (ii) competence plays as a direct determinant of performance; and (iii) competence can be an indirect influence of performance. Under educational circumstances, the relationship between competence and performance has been confirmed by previous research [12]. By conducting a questionnaire among first-year students studying psychology or educational sciences at Ghent University, Ferla et al. (2010) confirmed that students' self-perceived competence significantly impacts performance [31]. These findings were supported by Nwosu et al. (2018) [32] and Guo et al., (2021) [33].

2.3.2 Hypotheses development. Some studies have been conducted recently to examine the relationship between digital competence (or digital literacy) and core determinants (performance expectancy, effort expectancy, social influence, facilitating conditions) of continuance intention based on the UTAUT model. Nikou & Aavakare (2021) employed SEM and used Partial Least Squares (PLS) to investigate the impact of digital literacy on the intention to use digital technologies among Finnish university staff and students. They verified that digital literacy directly impacts staff and student's performance and effort expectations [34]. This research was supported by Jang et al. (2021), an international comparative study which employed SEM and multigroup analysis (MGA) for data analysis [35].

Based on the above literature, the researcher argues that employee digital competence may directly impact their expectancy concerning the effort required to learn and use new digital technology for working purposes (effort expectancy) and its impact on their job performance (performance expectancy). Also, the researcher assumes that digital competence affects an individual's perception regarding "important others believe he or she should use the new system" [28]. Given the above, this study hypothesizes the following:

H1a. Digital competence has a positive impact on performance expectancy

H1b. Digital competence has a positive impact on effort expectancy

H1c. Digital competence has a positive impact on social influence

Competence and self-efficacy are both key concepts in psychology but conceptualized differently, one at the level of the person (perceived competence) and one at the level of the behavior (self-efficacy) [36]. Regarding the relationship between digital competence and self-efficacy, Galindo-Domínguez & Bezanilla (2021) confirmed that digital competence is a good predictor of academic self-efficacy based on the survey results from 200 students from different degrees [37]. This finding was supported by Alex et al. (2021) [38], with a similar result from surveying 493 Indian teachers. Meanwhile, using the same research method, Hatlevik (2017) showed the opposite side of the research result when they demonstrated that basic ICT and online collaboration self-efficacy are two

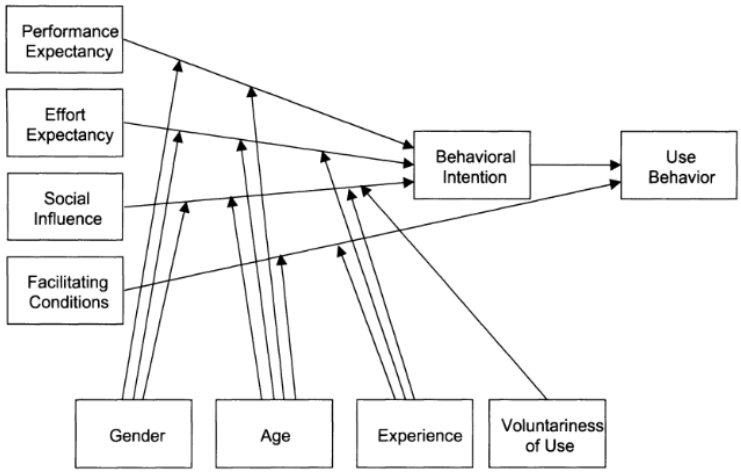


Figure 1: Unified Theory of Acceptance and Use of Technology (UTAUT), developed by Venkatesh et al., (2003) [28]

primary predictors of teachers' digital competence [39]. Based on the studies mentioned above, the following hypothesis is proposed:

H1d. Digital competence positively impacts non-academic staff's self-efficacy

Competence is positively related to an individual's personal accomplishments [40] and overall work-related performance [41]. According to Meske & Junglas (2021), the users' perception of competence when using digital tools in the work environment will also affect their performance [42]. Recently, the research conducted by Pacheco & Coello-Montecel, (2022), one of the latest research on this topic, demonstrated that digital competence is the catalyst for better performance with the mediating role of psychological empowerment [43]. By integrating digital literacy with the UTAUT model in education, Mohammadyari & Singh (2015) claim that digital literacy significantly impacts the intention to use e-learning and performance [44]. This relationship was then supported by Mehrvarz et al. (2021) when they surveyed over 300 students and verified that digital competence positively influences students' academic performance with the mediator role of informal digital learning [45]. Hence, the following hypothesis is proposed based on the studies mentioned above.

H1e. Digital competence positively impacts on job performance

Regarding the mediation effect of users' behavioral intention determinants, some studies indicate that these determinants positively mediate the relationship between digital competence (literacy) and intention to use digital technologies. Nikou & Aavakare (2021) verified that although there was no direct relationship between digital competence and intention to use digital technologies, this path was fully mediated through performance expectancy and effort expectancy [34]. In another comparative study, Jang et al. (2021) ran a mediation test, and they confirmed that while the path between digital literacy and intention to use was fully mediated by performance expectancy, there was no mediation role of effort expectancy in the correlation of these two variables [35]. Based on the studies mentioned above, the following hypothesis is proposed:

H2a. There is a significant impact of performance expectancy on intention to use digital technologies

H2b. There is a significant impact of effort expectancy on intention to use digital technologies

According to the attitude - social influence - self-efficacy (ASE) model (developed by Vries et al., 1998), social influence, attitude, and self-efficacy are the three proximal cognitive factors that directly impact intention and behavior. Social influence in the ASE model results from social norms, perceived behaviors of others, and direct pressure or support to perform a particular behavior [46]. Wang & Lin's (2011) research strengthened the relationship between social influence and behavioral intention. Based on the information system success model, the researchers developed a conceptual framework incorporating information quality, system quality, blog function quality (three types of blog platform qualities), and social influence as critical determinants of bloggers' usage intention. Via a web survey to collect the empirical data, the study's results revealed that social influence affects bloggers' usage intention directly and indirectly through blog flatform qualities [47]. However, unexpectedly and contrary to earlier studies, the result of the experimental research of Teo et al. (2019) showed no effect of social influence on consumers' purchase intention via Marketing on Instagram [48]. Hence, although the relationship between social influence and behavioral intention has received a long-standing interest from scholars, the research results are still inconsistent. Based on the above justification, this study proposes hypotheses H2c as follows:

H2c. There is a significant impact of social influence on continuance intention to use digital technologies for work

Self-efficacy is considered an indirect construct in the UTAUT model [28]. However, some researchers indicated that not considering attitude and self-efficacy as direct determinants of intention is a limitation of the UTAUT model. According to Moghavvemi (2015), self-efficacy should be an antecedent of behavioral intention. After combining the UTAUT model with the Entrepreneurial Potential Model (Krueger & Brazeal, 1994), the author verified that perceived feasibility (self-efficacy) positively impacts intention to use [49]. At the same time, Veer et al. (2015) examined the effect of performance expectancy, effort expectance, social influence and

self-efficacy on the intention to use e-Health applications among older people. The results revealed that self-efficacy was related to the intention to use [50]. Based on the studies mentioned above, the following hypothesis is proposed:

H2d. There is a significant impact of self-efficacy on continuance intention to use digital technologies for work

Lee & Ko (2010) conducted research in hospitals in seven provinces in Korea to test the effect of individual-level variables (including self-efficacy, job position, years of experience, employment status, and positive affectivity) and group-level variables on nursing performance. They concluded that individual-level variables, including self-efficacy, are positively related to nursing performance [51]. In the educational context, the association between self-efficacy and performance among students has been confirmed by some scholars. To examine the mediating effect of metacognitive in the association of academic self-efficacy and academic performance, Hayat et al. (2020) surveyed 279 medicine students. The results denoted that academic self-efficacy, directly and indirectly, affects academic performance (through the mediation of metacognitive learning) [52]. Similarly, Talsma et al. (2021) once again demonstrated the impact of self-efficacy on academic performance among first-year university students [53]. Against this background, with the support of empirical evidence, this study postulates a significant relationship between self-efficacy and non-academic staff's job performance.

H2e. There is a significant impact of self-efficacy on job performance among non-academic staff

The connection between continuance intention and performance has been demonstrated in Mohammadyari & Singh's (2015) research with New Zealand accountants as participants. By integrating the concept of digital literacy and the UTAUT constructs, the researchers denoted that the relationship between the intention to continue using e-learning of accountants and their performance is significant [44]. This finding had verified by Marsh (2018) when she conducted research among 142 British employees to understand the effect of digital literacy on employees' continuance intentions and performance. The study denoted that information system continuance intention impacts job performance [54]. In education, Maqableh et al. (2021) researched the determinants of university students' academic performance with the mediating role of Internet usage continuance intention. The data collected for 476 students and analyzed by SPSS and AMOS show that continuance intention, one of the four determinants, positively influences academic performance [55]. Based on these above arguments, non-academic staff's intention to use digital technologies for work is expected to affect their job performance. Therefore, the following hypothesis is proposed to be tested:

H3. Continuance intention to use digital technologies for work positively affects job performance

2.3.3 Conceptual Framework of the Study. Based on the above discussion and established hypotheses, a conceptual framework is proposed to examine the impact of the digital competence of non-academic staff on their job performance in Figure 2. The UTAUT model is the best alternative to develop a conceptual framework for the study for the following reasons. First, it has been argued that the UTAUT model is the most widely used technology acceptance model to predict the behavioral intention to use digital technologies Jang et al. (2021), which is demonstrated to be relevant to digital competence (literacy) [34, 35, 56]. Second, this model has been recently used in the context of digital transformation [44], especially in education settings [57]. Based on the above justification, developing the conceptual framework based on the UTUAT model is reasonable.

However, adopting the UTAUT model is contextualized in its adaptation based on the study's focus, especially on the continuance intention to use digital technology. For example, to assess the impact of information literacy and digital literacy on the intention to use digital technologies among Finnish university staff and students, Nikou & Aavakare (2021) adopted only two determinants in the UTAUT model, performance expectancy and effort expectancy, and verified the mediating role of these two factors. The researchers clearly justified choosing only these two variables [34]. Based on the study's context, usage behavior has been excluded from the model because the research only focused on intention behavior rather than actual usage.

In the current study, the UTAUT model is adapted to fit with the focus on the impact of digital competence on job performance. Three out of four direct determinants of the acceptance and usage of digital technologies, including performance expectancy, effort expectance, and social influence, will be adopted from the UTAUT mode. Facilitating conditions are excluded because it is an external variable unrelated to the continuance intention to use digital technologies at the individual level. Moreover, this study will investigate the continuance intention rather than the usage of digital technologies; thus, the usage behavior is excluded. Based on the discussion in the previous part, the relationship of all the above hypotheses is illustrated below:

3 PROPOSED METHODOLOGY

In this study, the quantitative research methodology with a survey design will be applied in the current study because of its advantages. Firstly, survey design, considered the prevalent method for quantitative research in social science, is an excellent method to measure unobservable data. Secondly, survey research is well suited for gathering information remotely from a large population. Thirdly, respondents tend to prefer to participate in a questionnaire survey because it lets them respond at their convenience. Fourthly, questionnaire surveys allow the researcher to reach potential participants without a sampling frame. Finally, compared to other methods, survey research is economical in terms of researcher time, effort, and cost [59].

In the current study, non-academic staff, including managers, training officers, student affairs officers, administrative officers, IT men, admission officers, librarians, quality assurance officers, building operations and maintenance officers, corporate relations officers, public relations officers, curriculum developers, human resources officers, and accountants in the Vietnam university, are set as participants. Other job positions, such as security staff, transportation workers and catering staff will not include since their job characteristics do not require to use digital technologies frequently. The study locations are four university campuses, including Hanoi, Da Nang, Ho Chi Minh, and Can Tho, corresponding to the North, Middle, and South parts of Vietnam.

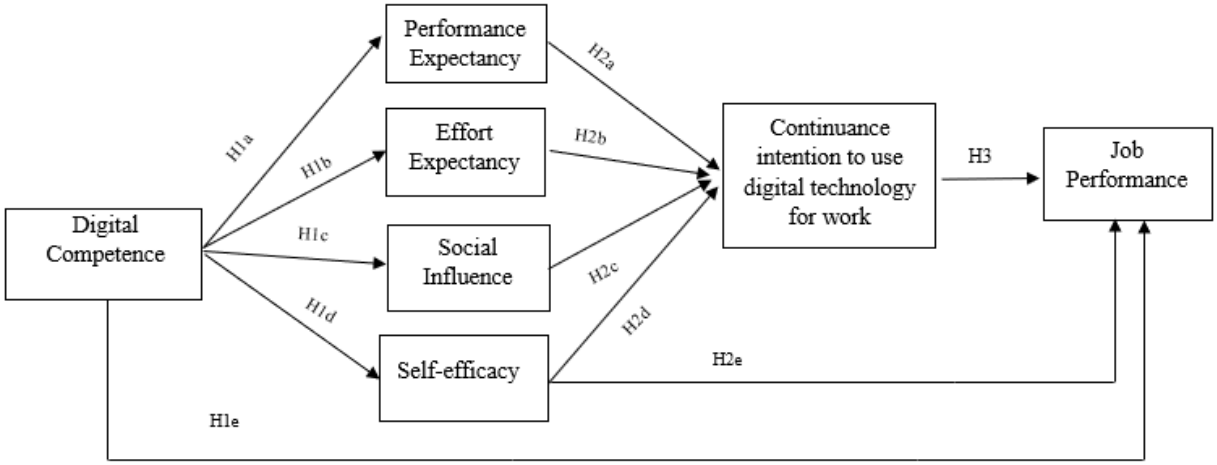


Figure 2: Conceptual Framework of Study Adapted from the UTAUT model, developed by Venkatesh et al. (2003)

Due to the large population and the confidentiality and sensitivity of human resource data, developing a "sampling frame" of non-academic staff in a large educational organization is nearly impossible. For this reason, the researcher will employ a snowball sampling technique to reach potential participants through social media and internal email groups. The online survey will be conducted via Google form because it is a free online tool that allows the researcher to collect data quickly and efficiently.

First of all, the researcher will select initial participants who are called "seeds" that meet the following requirements: (i) They are non-teaching staff at an educational organization; (ii) They have a vast network of prospective participants; (iii) They are willing to participate in the research and introduce the new cases. The researcher will then contact the initial participant to introduce the research purposes, processes, and data selection procedures. After the selection process, the six initial seeds will be invited to complete the questionnaires and guided to share with other cases. The process will be continued until the researcher gets 250 samples.

4 DISCUSSION

Notwithstanding the central role of digital competence in society and organizations in the digital age, international surveys and academic literature reveal that global workers are facing a digital skills crisis. Vietnam, a developing nation in Southeast Asia, saw a similar pattern, and even it is more apparent that the working force in Vietnam lacks digital proficiency [7]. Much research in the educational section revealed that teachers' and students' digital competence levels were maintained at a basic [20] or an intermediate level [60]. In this context, limited research focuses on the digital competence of non-academic staff, who play a vital role in educational institutions, in the literature; hence, little is known about non-academic staff"s capability in applying and utilizing digital technologies around them in work.

This study expects that the determinants of intention to use digital technologies (performance expectance, effort expectance, and social influence) will mediate the relationship between digital competence and job performance. Also, the research will be expected that digital competence will significantly impact job performance directly and indirectly through self-efficacy. The findings of this study may assist educational organizations in selecting the most appropriate strategy for improving digital competence for non-academic staff by better understanding its impact on job performance. The results from this study also drive non-academic staff to increase their acceptance and usage of digital technologies for work; as a result, increasing their job performance. On the other hand, this study may contribute to the knowledge of digital transformation in general and digital competence in particular. Based on the significance, conducting this study in the context of higher education in Vietnam is both timely and critical.

REFERENCES

[1] Murawski, M., & Bick, M. (2017). Digital competences of the workforce – a research topic? *Business Process Management Journal*, 23(3), 721–734. https://doi.org/10.1108/BPMJ-06-2016-0126
[2] Benavides, L. M. C., Tamayo Arias, J. A., Arango Serna, M. D., Branch Bedoya, J. W., & Burgos, D. (2020). Digital Transformation in Higher Education Institutions: A Systematic Literature Review. *Sensors*, 20(11), Article 11. https://doi.org/10.3390/s20113291
[3] Kane, G. (2019). The Technology Fallacy: People Are the Real Key to Digital Transformation. *Research-Technology Management*, 62(6), 44–49. https://doi.org/10.1080/08956308.2019.1661079
[4] Anusca Ferrari. (2013). DIGCOMP: A framework for developing and understanding digital competence in Europe. *Publications Office*. https://data.europa.eu/doi/10.2788/52966
[5] British Council Viet Nam. (2022). Readiness of digital transformation in Vietnamese universities. britishcouncil.vn/sites/default/files/report_readiness_of_digital_transformation_in_vietnamese_universities.pdf
[6] European court of auditors. (2021). EU actions to address low digital skills.
[7] Trinh. (2022). Phát trien nang luc so cho luc luong lao dong. *Nhandan*.
[8] Kundi, Y. M., Aboramadan, M., Elhamalawi, E. M. I., & Shahid, S. (2021). Employee psychological well-being and job performance: Exploring mediating and moderating mechanisms. *International Journal of Organizational Analysis*, 29(3), 736–754. https://doi.org/10.1108/IJOA-05-2020-2204
[9] Bolli, T., & Pusterla, F. (2022). Decomposing the effects of digitalization on workers' job satisfaction. International Review of Economics, 69(2), 263–300. https://doi.org/10.1007/s12232-022-00392-6
[10] Naim, M. F., & Lenka, U. (2017). The impact of social media and collaboration on Gen Y employees' engagement. *International Journal of Development Issues*, 16(3),

289–299. https://doi.org/10.1108/IJDI-04-2017-0041
[11] Koopmans, L., Bernaards, C. M., Hildebrandt, V. H., Schaufeli, W. B., de Vet Henrica, C. W., & van der Beek, A. J. (2011). Conceptual Frameworks of Individual Work Performance: A Systematic Review. *Journal of Occupational & Environmental Medicine*, 53(8), 856–866. https://doi.org/10.1097/JOM.0b013e318226a763
[12] Campbell, J. P., & Wiernik, B. M. (2015). The Modeling and Assessment of Work Performance. *Annual Review of Organizational Psychology and Organizational Behavior*, 2(1), 47–74. https://doi.org/10.1146/annurev-orgpsych-032414-111427
[13] Bonache, J., & Noethen, D. (2014). The impact of individual performance on organizational success and its implications for the management of expatriates. *The International Journal of Human Resource Management*, 25(14), 1960–1977. https://doi.org/10.1080/09585192.2013.870287
[14] Vosloban, R. I. (2012). The Influence of the Employee's Performance on the Company's Growth—A Managerial Perspective. *Procedia Economics and Finance*, 3, 660–665. https://doi.org/10.1016/S2212-5671(12)00211-0
[15] Scotter, J. R. V. (2000). Relationships of Task Performance and Contextual Performance with Turnover, Job Satisfaction, and Affective Commitment. *Human Resource Management Review*, 10(1), 79–95. https://doi.org/10.1016/S1053-4822(99)00040-6
[16] Alexander, N. A., Jang, S. T., & Kankane, S. (2017). The Performance Cycle: The Association between Student Achievement and State Policies Tying Together Teacher Performance, Student Achievement, and Accountability. *American Journal of Education*, 123(3), 413–446. https://doi.org/10.1086/691229
[17] European Commission. (2006). Recommendation of the European Parliament and of the Council. https://eur-lex.europa.eu/legal-content/en/TXT/?uri$=$CELEX:32006H0962&qid$=$1496720114366
[18] Anusca Ferrari. (2012). Digital competence in practce: An analysis of frameworks. *Publications Office*. https://data.europa.eu/doi/10.2791/82116
[19] Bloom, B. S. (1956). Taxonomy of Educational Objectives, Handbook: The Cognitive Domain. David McKay. New York.
[20] Zhao, Y., Pinto Llorente, A. M., & Sánchez Gómez, M. C. (2021). Digital competence in higher education research: A systematic literature review. *Computers & Education*, 168, 104212. https://doi.org/10.1016/j.compedu.2021.104212
[21] Stephanie Carretero, Riina Vuorikari, & Yves Punie. (2017). The Digital Competence Framework for Citizens.
[22] Campbell, J. P. (1990). Modeling the performance prediction problem in industrial and organizational psychology. In Handbook of industrial and organizational psychology, Vol. 1, 2nd ed (pp. 687–732). *Consulting Psychologists Press*.
[23] Guzmán-Ortiz, C. V., Navarro-Acosta, N. G., Florez-Garcia, W., & Vicente-Ramos, W. (2020). Impact of digital transformation on the individual job performance of insurance companies in Peru. *International Journal of Data and Network Science*, 337–346. https://doi.org/10.5267/j.ijdns.2020.9.005
[24] Roe, R. A. (1999). Work performance: A multiple regulation perspective. In International review of industrial and organizational psychology 1999, Vol. 14 (pp. 231–335). John Wiley & Sons Ltd.
[25] Pandey, J. (2018). Factors affecting job performance: An integrative review of literature. *Management Research Review*, 42(2), 263–289. https://doi.org/10.1108/MRR-02-2018-0051
[26] Sonnentag, S., & Frese, M. (2005). Performance Concepts and Performance Theory. In S. Sonnentag (Ed.), Psychological Management of Individual Performance (pp. 1–25). *John Wiley & Sons*, Ltd. https://doi.org/10.1002/0470013419.ch1
[27] Al-Ahmadi, H. (2009). Factors affecting performance of hospital nurses in Riyadh Region, Saudi Arabia. *International Journal of Health Care Quality Assurance*, 22(1), 40–54. https://doi.org/10.1108/09526860910927943
[28] Venkatesh, Morris, Davis, & Davis. (2003). User Acceptance of Information Technology: Toward a Unified View. MIS Quarterly, 27(3), 425. https://doi.org/10.2307/30036540
[29] Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review*, 84(2), 191–215. https://doi.org/10.1037/0033-295X.84.2.191
[30] Wang, C.-H., Shannon, D. M., & Ross, M. E. (2013). Students' characteristics, self-regulated learning, technology self-efficacy, and course outcomes in online learning. *Distance Education*, 34(3), 302–323. https://doi.org/10.1080/01587919.2013.835779
[31] Ferla, J., Valcke, M., & Schuyten, G. (2010). Judgments of self-perceived academic competence and their differential impact on students' achievement motivation, learning approach, and academic performance. *European Journal of Psychology of Education*, 25(4), 519–536. https://doi.org/10.1007/s10212-010-0030-9
[32] Nwosu, J. C., John, H. C., Izang, A. A., & Akorede, O. J. (2018). Assessment of Information and Communication Technology (ICT) Competence and Literacy Skills among Undergraduates as a Determinant Factor of Academic Achievement. *Educational Research and Reviews*, 13(15), 582–589.
[33] Guo, J., Chen, Z., & Zheng, B. (2021). Postgraduate Competence and Academic Research Performance: The Mediating Role of Psychological Capital. *Sustainability*, 13(11), Article 11. https://doi.org/10.3390/su13116469
[34] Nikou, S., & Aavakare, M. (2021). An assessment of the interplay between literacy and digital Technology in Higher Education. *Education and Information Technologies*, 26(4), 3893–3915. https://doi.org/10.1007/s10639-021-10451-0
[35] Jang, M., Aavakare, M., Nikou, S., & Kim, S. (2021). The impact of literacy on intention to use digital technology for learning: A comparative study of Korea and Finland. *Telecommunications Policy*, 45(7), 102154. https://doi.org/10.1016/j.telpol.2021.102154
[36] Rodgers, W. M., Markland, D., Selzler, A.-M., Murray, T. C., & Wilson, P. M. (2014). Distinguishing Perceived Competence and Self-Efficacy: An Example From Exercise. *Research Quarterly for Exercise and Sport*, 85(4), 527–539. https://doi.org/10.1080/02701367.2014.961050
[37] Galindo-Domínguez, H., & Bezanilla, M.-J. (2021). Promoting Time Management and Self-Efficacy Through Digital Competence in University Students: A Mediational Model. *Contemporary Educational Technology*, 13(2), ep294. https://doi.org/10.30935/cedtech/9607
[38] Alex, D., Varghese, J., & Thomas, M. (2021). Embracing Digitalisation—Exploring the Relationship Between Digital Competence and Self-Efficacy of School Teachers. 7.
[39] Hatlevik, O. E. (2017). Examining the Relationship between Teachers' Self-Efficacy, their Digital Competence, Strategies to Evaluate Information, and use of ICT at School. *Scandinavian Journal of Educational Research*, 61(5), 555–567. https://doi.org/10.1080/00313831.2016.1172501
[40] Broeck, A., Vansteenkiste, M., Witte, H., Soenens, B., & Lens, W. (2010). Capturing autonomy, competence, and relatedness at work: Construction and initial validation of the Work-related Basic Need Satisfaction scale. *Journal of Occupational and Organizational Psychology*, 83(4), 981–1002. https://doi.org/10.1348/096317909X481382
[41] Stajkovic, A. D., & Luthans, F. (1998). Self-efficacy and work-related performance: A meta-analysis. *Psychological Bulletin*, 124(2), 240–261. https://doi.org/10.1037/0033-2909.124.2.240
[42] Meske, C., & Junglas, I. (2021). Investigating the elicitation of employees' support towards digital workplace transformation. *Behaviour & Information Technology*, 40(11), 1120–1136. https://doi.org/10.1080/0144929X.2020.1742382
[43] Pacheco, P. O., & Coello-Montecel, D. (2023). Does psychological empowerment mediate the relationship between digital competencies and job performance? *Computers in Human Behavior*, 140, 107575. https://doi.org/10.1016/j.chb.2022.107575
[44] Mohammadyari, S., & Singh, H. (2015). Understanding the effect of e-learning on individual performance: The role of digital literacy. *Computers & Education*, 82, 11–25. https://doi.org/10.1016/j.compedu.2014.10.025
[45] Mehrvarz, M., Heidari, E., Farrokhnia, M., & Noroozi, O. (2021). The mediating role of digital informal learning in the relationship between students' digital competence and their academic performance. *Computers & Education*, 167, 104184. https://doi.org/10.1016/j.compedu.2021.104184
[46] De Vries, H., Mudde, A. N., Dijkstra, A., & Willemsen, M. C. (1998). Differential Beliefs, Perceived Social Influences, and Self-Efficacy Expectations among Smokers in Various Motivational Phases. *Preventive Medicine*, 27(5), 681–689. https://doi.org/10.1006/pmed.1998.0344
[47] Wang, S., & Chuan-Chuan Lin, J. (2011). The effect of social influence on bloggers' usage intention. *Online Information Review*, 35(1), 50–65. https://doi.org/10.1108/14684521111113588
[48] Teo, L. X., Leng, H. K., & Phua, Y. X. P. (2019). Marketing on Instagram: Social influence and image quality on perception of quality and purchase intention. *International Journal of Sports Marketing and Sponsorship*, 20(2), 321–332. https://doi.org/10.1108/IJSMS-04-2018-0028
[49] Moghavvemi, S. (2015). Impact of Perceived Self-Efficacy and Capability to Use IT Innovation on Individual Use Behaviour. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.2561739
[50] de Veer, A. J. E., Peeters, J. M., Brabers, A. E., Schellevis, F. G., Rademakers, J. J. J., & Francke, A. L. (2015). Determinants of the intention to use e-Health by community dwelling older people. *BMC Health Services Research*, 15(1), 103. https://doi.org/10.1186/s12913-015-0765-8
[51] Lee, T. W., & Ko, Y. K. (2010). Effects of self-efficacy, affectivity and collective efficacy on nursing performance of hospital nurses. *Journal of Advanced Nursing*, 66(4), 839–848. https://doi.org/10.1111/j.1365-2648.2009.05244.x
[52] Hayat, A. A., Shateri, K., Amini, M., & Shokrpour, N. (2020). Relationships between academic self-efficacy, learning-related emotions, and metacognitive learning strategies with academic performance in medical students: A structural equation model. *BMC Medical Education*, 20(1), 76. https://doi.org/10.1186/s12909-020-01995-9
[53] Talsma, K., Robertson, K., Thomas, C., & Norris, K. (2021). COVID-19 Beliefs, Self-Efficacy and Academic Performance in First-year University Students: Cohort Comparison and Mediation Analysis. *Frontiers in Psychology*, 12. https://www.frontiersin.org/articles/10.3389/fpsyg.2021.643408
[54] Marsh, E. (2018). Understanding the Effect of Digital Literacy on Employees' Digital Workplace Continuance Intentions and Individual Performance: *International Journal of Digital Literacy and Digital Competence*, 9(2), 15–33. https://doi.org/10.4018/IJDLDC.2018040102
[55] Maqableh, M., Jaradat, M., & Azzam, A. (2021). Exploring the determinants of students' academic performance at university level: The mediating role of internet usage continuance intention. *Education and Information Technologies*, 26(4), 4003–4025. https://doi.org/10.1007/s10639-021-10453-y

[56] He, T., & Li, S. (2019). A comparative study of digital informal learning: The effects of digital competence and technology expectancy. *British Journal of Educational Technology*, 50(4), 1744–1758. https://doi.org/10.1111/bjet.12778

[57] Thomas, T. D., Singh, L., & Gaffar, K. (2013). The utility of the UTAUT model in explaining mobile learning adoption in higher education in Guyana. *International Journal of Education and Development Using Information and Communication Technology* (IJEDICT), 9(3), 71–85. Scopus.

[58] Creswell, J. W. (2009). Research design: Qualitative, quantitative, and mixed methods approaches, 3rd ed (pp. xxix, 260). *Sage Publications*, Inc.

[59] Bhattacherjee, A. (2012). Social science research: Principles, methods, and practices (Second edition). *Anol Bhattacherjee*.

[60] Martzoukou, K., Fulton, C., Kostagiolas, P., & Lavranos, C. (2020). A study of higher education students' self-perceived digital competences for learning and everyday life online participation. *Journal of Documentation*, 76(6), 1413–1458. https://doi.org/10.1108/JD-03-2020-0041